

IEEE VR



SAINT-MALO, FRANCE
March 8-12, 2025

Reviewing and Publication Models in the Era of Online Publishing and Generative AI

J. Edward Swan II

Mississippi State University

Saturday, 8 March 2025

Are you happy with the way that peer review is currently implemented?

Outline

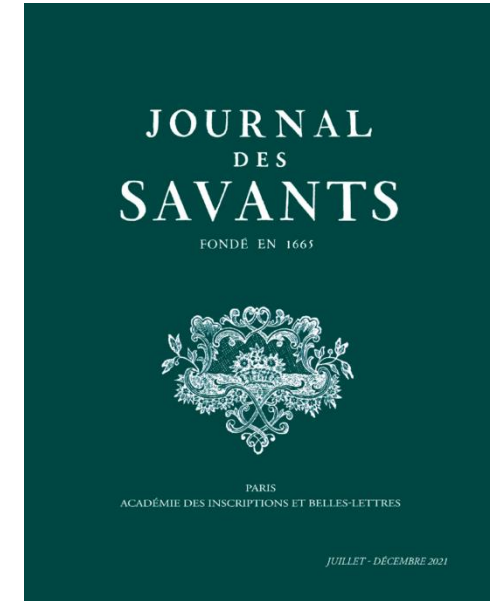
- **A brief history of scholarly publishing and reviewing models**
- **Consequences of paper-based publishing models**
- **What is our reviewing model?**
- **Alternative reviewing models**
- **Alternative publication models**
- **A more radical reviewing and publication model**
- **Potential benefits of new models**
- **Concluding thoughts**

A brief history of scholarly publishing and reviewing models

- **A brief history of scholarly publishing and reviewing models**
- **Consequences of paper-based publishing models**
- **What is our reviewing model?**
- **Alternative reviewing models**
- **Alternative publication models**
- **A more radical reviewing and publication model**
- **Potential benefits of new models**
- **Concluding thoughts**

Beginnings of scholarly publishing

- Founding of learned societies
 - Compagnie du Gai Sçavoir, 1323
 - Royal Society of London, 1660
- Movable-type printing press invented
 - Johannes Gutenberg, 1440
- Publishing of scholarly journals
 - Begun in the 1600's; expanded greatly in 1800's
 - Journal des Sçavans (January 1665)
 - Philosophical Transactions of the Royal Society (March 1665)
 - Published Bulletins, Proceedings, Transactions



Decision methods and reviewing

- **Beginning (1660): a single editor solicited and selected papers**
 - **Philosophical Transactions of the Royal Society: Henry Oldenburg**
- **Team of editors (early-to-mid 1800's)**
 - **Additional expertise**
 - **Voting on what to publish**
 - **Began to seek reports of referees**
- **Sharing referee suggestions with authors, guiding author response**
 - **George Gabriel Stokes, secretary of the Royal Society, 1854–85**

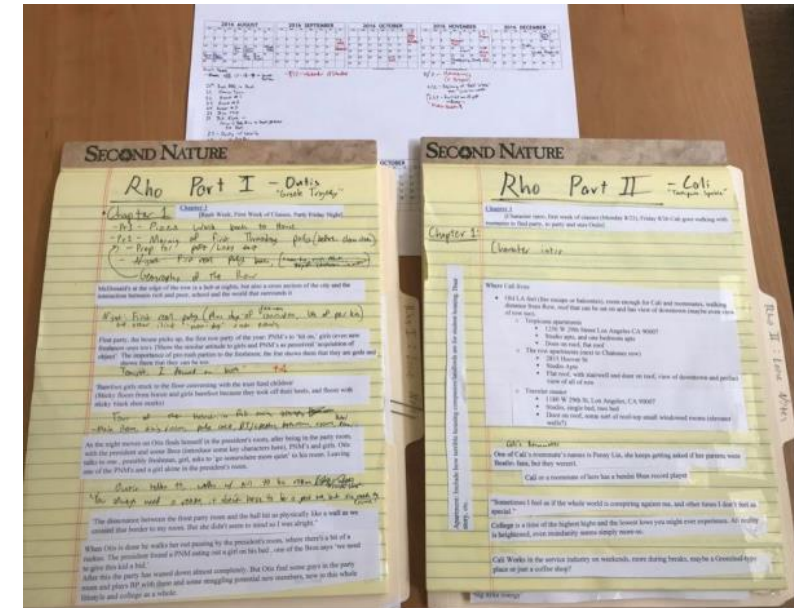
Peer Review: Not as old as you might think

- Oxford English Dictionary
 - “a form of review of competence by others in the same occupation”
 - first used in 1967
- What does it mean?
 - Modern connotation: **proof beyond reasonable doubt**
- Began to be used by grant-making bodies (1970's)
- Historians of scholarly publishing ask:
 - Is **peer review** up to the challenges of the online age?
 - **Not a sacred cow**, but the currently dominant practice in a long and varied history of reviewing practices



Paper-based scholarly publishing (1960's–90's)

- Conduct a research project
 - Write code, evaluate, perhaps empirical experiment with human subjects, etc.
- Write paper
- Mail 4+ copies to conference (journal)
 - Graphics or photos? Glue color photos onto pages
 - Late-night run to FedEx
- Conference mails reviewers, reviewers mail conference, conference mails author
- Revisions, copyediting, proofing by mail
- Journal typesets issue; prints; mails to subscribers and libraries
- Scholar: receive issue in the mail. Otherwise, visit library.



e.g., [Nosek, 2024]

Consequences of paper-based publishing models

- A brief history of scholarly publishing and reviewing models
- **Consequences of paper-based publishing models**
- What is our reviewing model?
- Alternative reviewing models
- Alternative publication models
- A more radical reviewing and publication model
- Potential benefits of new models
- Concluding thoughts

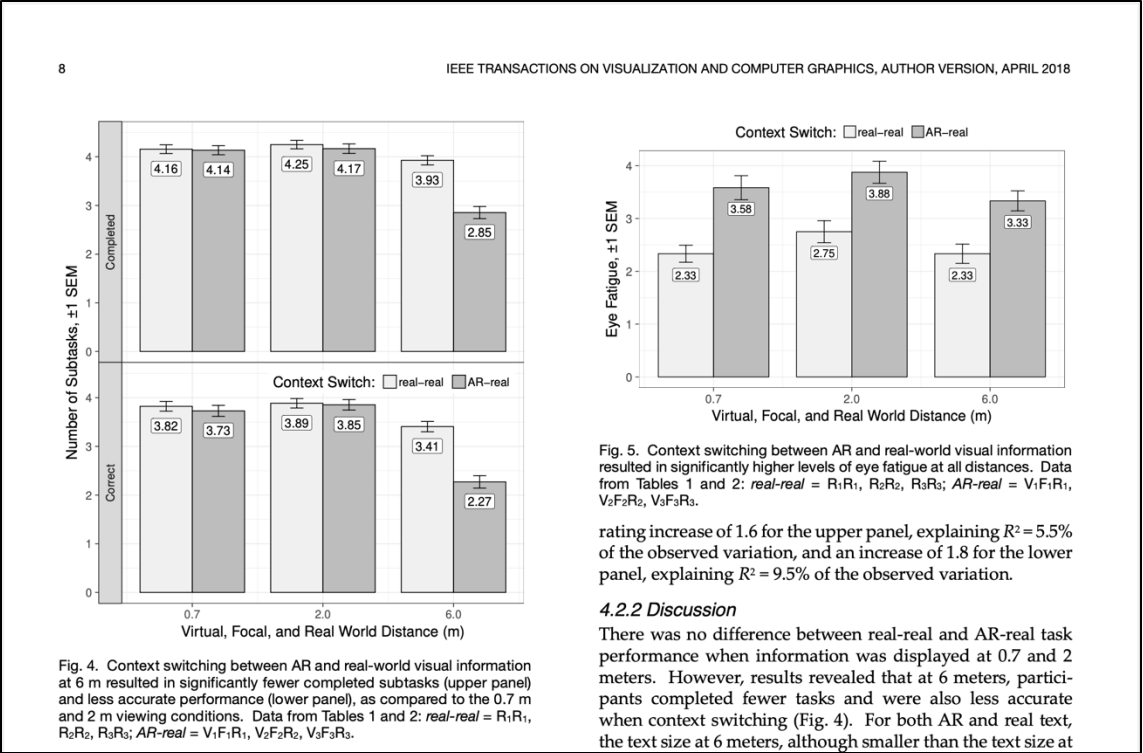
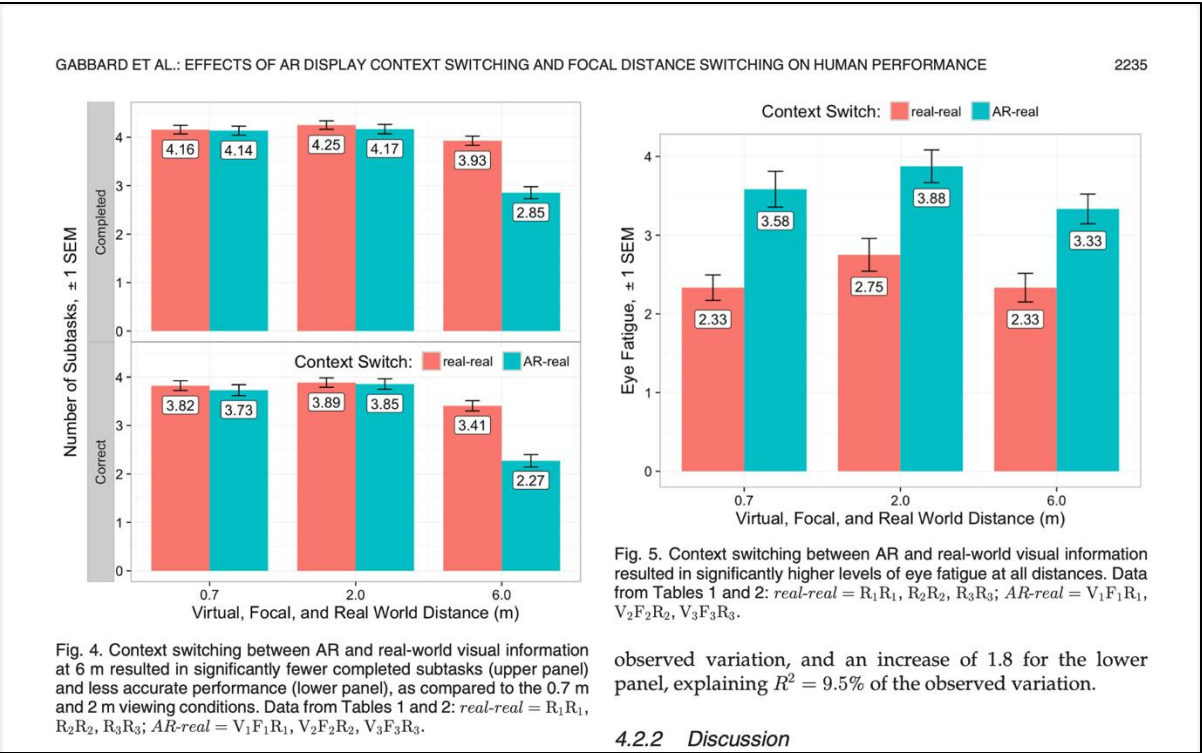
Consequences of paper-based publishing

- Publisher played useful role
 - Manage peer review
 - Print and distribute proceedings / journal
- Peer review only happens when research is complete
- Only paper is reviewed
 - Not research plan, motivation, code, data, materials, statistical analysis, etc.
- Only reviewed by a few people, somewhat selected at random
- Review results in binary decision: **accept** (maybe with edits) or **reject**
- Review and decision process is **opaque** and **unknowable**
- Decision is **permanent** and **unrevisable**
- Decision often based on unknowable facts
 - Whether or not the research should have been conducted in the first place
 - **Importance** or **impact** of research, as seen 5+ / 10+ years in the future

e.g., [Nosek, 2024]

Cannot update archived paper

- JL Gabbard, DG Mehra, and JE Swan II. Effects of AR Display Context Switching and Focal Distance Switching on Human Performance. *IEEE Transactions on Visualization and Computer Graphics*, 25(6):2228–2241, May 2018. DOI: 10.1109/TVCG.2018.2832633.

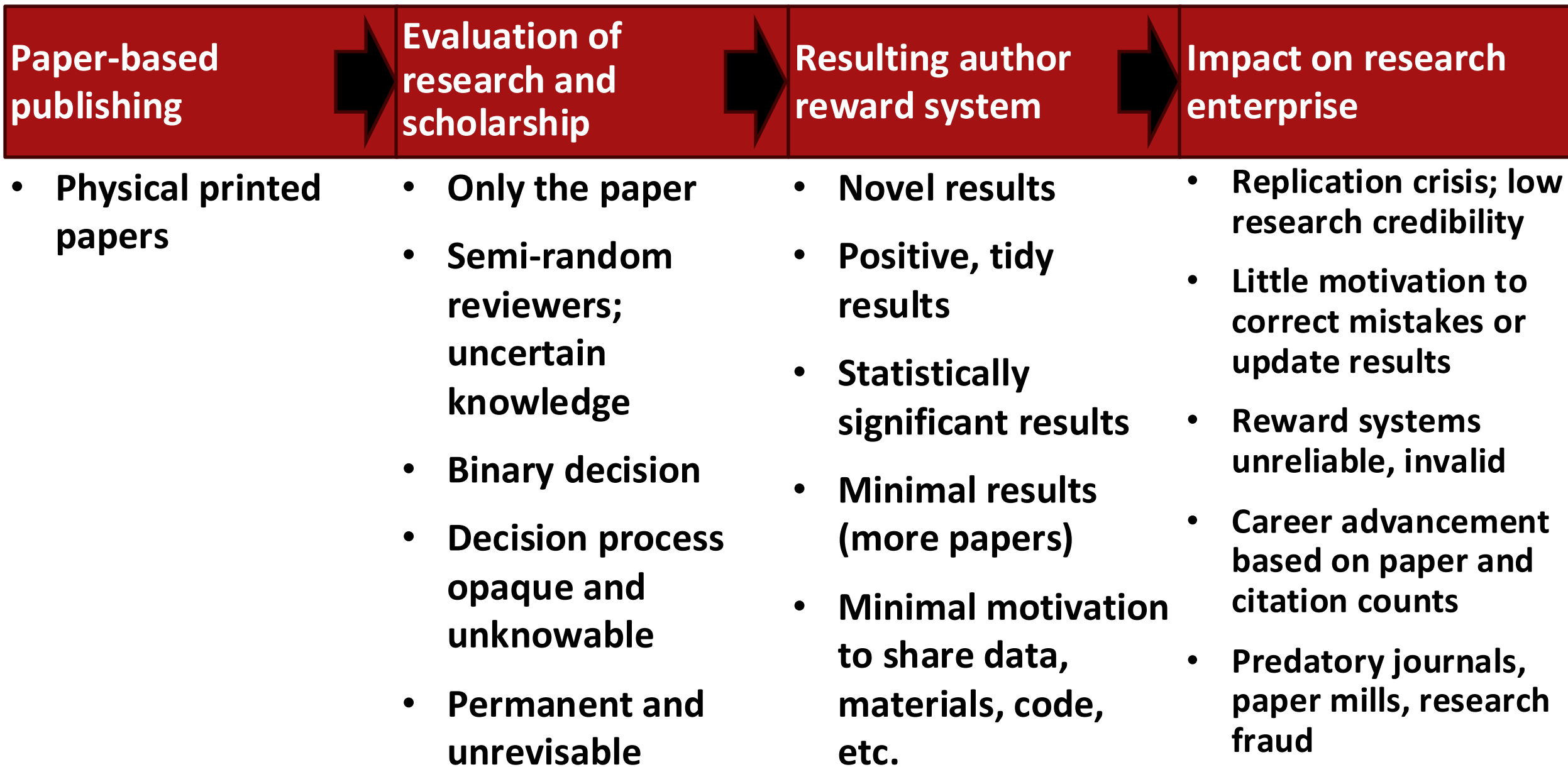


Digital Library Version:

<https://ieeexplore.ieee.org/document/8353823>

Author Version on Website:

<https://ed-swan.github.io/index.html>



But, now it is digital! Why does scholarly reviewing and publishing still have these problems?

- Risk and uncertainty
- Inertia; system justification
- Publisher business models
 - Important inputs provided for free: **articles**, **peer reviews**
 - Publishers no longer print physical proceedings → run digital libraries
 - Inelastic demand for articles
 - 40% margins → **serials crisis**
- However, much energy devoted to developing alternative reviewing and publishing models
 - Open Access
 - Shadow Libraries (Pirate Libraries)
 - Open Science

What is our reviewing model?

- A brief history of scholarly publishing and reviewing models
- Consequences of paper-based publishing models
- **What is our reviewing model?**
- Alternative reviewing models
- Alternative publication models
- A more radical reviewing and publication model
- Potential benefits of new models
- Concluding thoughts

Our implementation of decision and peer review (VR 2025)

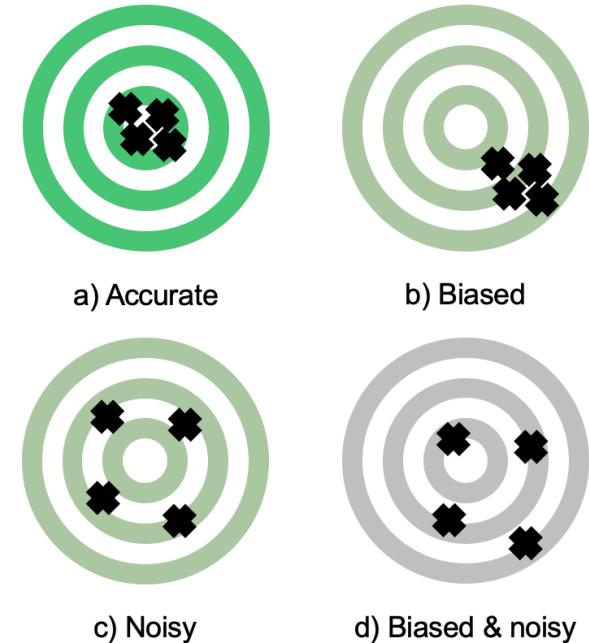
- **Stage 1:** Paper receives primary and secondary reviewer from IPC
 - **Desk rejections:** not anonymized, not formatted, out of scope, lacking IRB for human subject studies
 - **Early rejections:**
 - Secondary reviewer performs full review
 - If low score, primary performs full review, and if consensus early rejection
- **Stage 2:** Primary and secondary recruit two external reviewers
 - External reviews + secondary review = **3 reviews + scores**
 - **Primary leads online discussion** → (1) conditional accept as paper [don't decide TVCG / Conference], (2) major revision, (3) conditional accept as poster, (4) reject, (5) no consensus
- **Stage 3:** Supercommittee discussion of papers
 - **Live meetings** → (1) recommended for conditional acceptance, (2) poster, (3) reject, (4) not recommended but high reviewer scores
 - Authors in category 4 given opportunity to **rebut**
 - **Final decision, including TVCG / Conference**
- **Stage 4:** Minor revision cycle, overseen by primary

Alternative reviewing models

- A brief history of scholarly publishing and reviewing models
- Consequences of paper-based publishing models
- What is our reviewing model?
- **Alternative reviewing models**
- Alternative publication models
- A more radical reviewing and publication model
- Potential benefits of new models
- Concluding thoughts

Reviews completed independently

- VR conference reviews influence each other
 - Discussion phase, reviewers can alter their score
 - Reviews are initially written independently, but can be circumvented
- Alternative method: **reviews completed and scored independently**
 - Hidden Brain interview with **Daniel Kahneman (2021)** → if human judgements independent, even if noisy, average judgement less biased than any single judgement
- Example journal: **Transactions on Visualization and Computer Graphics**
 - Solicit 3 external reviews
 - If the 3 reviews agree:
 - If positive, accept (or minor revision)
 - If negative, reject
 - If 2 reviewers agree, and 1 disagrees:
 - If majority positive, accept (or major or minor revision)
 - If majority negative, reject (or possibly major revision)
 - This method implicitly averages the reviews



[Hidden Brain, 2021;
Kahneman et al., 2021;
Surowiecki, 2005]

Single- vs double-blind reviewing

- VR conference reviews double-blind
 - **Advantage:** reduction in reviewer bias
 - **Disadvantages:**
 - Time-consuming and difficult to properly anonymize a paper
 - If not properly anonymized, implied moral failing
 - If non-archival previous publication (poster abstract, thesis, arXiv, etc.), anonymity difficult
 - Motivated reviewers can almost always break anonymity
 - Particularly difficult for series of replicate + extend empirical studies
- Most journals are single-blind
- **Hybrid approach:** Transactions on Visualization and Computer Graphics
 - Single-blind is default, but double-blind submission is possible

Non-anonymous reviewers

SYSTEMATIC REVIEW article

Front. Robot. AI , 16 April 2018

Sec. Virtual Environments

Volume 5 - 2018 | <https://doi.org/10.3389/frobt.2018.00037>

A Systematic Review of 10 Years of Augmented Reality Usability Studies: 2005 to 2014



Arindam Dey^{1*}



Mark Billinghamurst¹



Robert W. Lindeman²



J. Edward Swan II³

¹ Empathic Computing Laboratory, University of South Australia, Mawson Lakes, SA, Australia

² Human Interface Technology Lab New Zealand (HIT Lab NZ), University of Canterbury, Christchurch, New Zealand

³ Mississippi State University, Starkville, MS, United States

Augmented Reality (AR) interfaces have been studied extensively over the last few decades, with a growing number of user-based experiments. In this paper, we systematically review 10 years of the most influential AR user studies, from 2005 to 2014. A total of 291 papers with 369 individual user studies have been reviewed and classified based on their application areas. The primary contribution of the review is to present the broad landscape of user-based AR research, and to provide a high-level view of how that landscape has changed. We summarize the high-level contributions from each category of papers, and present examples of the most influential user studies. We also identify areas where there have been few user studies, and opportunities for future research. Among other things, we find that there is a growing trend toward handheld AR user studies, and that most

Download article ▾

51,7K

Total views

8,9K

Downloads

341

Citations



View article impact >



View altmetric score >

Share on



Edited by



Doug A. Bowman

Virginia Tech, United States

Reviewed by



John Quarles

University of Texas at San Antonio, United States



George Papagiannakis

University of Crete, Greece

Public reviews

Superstoichiometric reversible and manipulable copper-ion intercalation in niobium selenide

[Yuanhe Sun](#), [Rui Qi](#), [Zhipeng Xue](#), [Qi Lei](#), [Yuanxin Zhao](#), [Zhiguo Ren](#), [Wei Zhang](#), [Jingying Si](#), [Haitao Li](#), [Yi Gao](#), [Wen Wen](#), [Xiaolong Li](#) ✉ & [Daming Zhu](#) ✉

[Nature Communications](#) **16**, Article number: 2099 (2025) | [Cite this article](#)

[Metrics](#)

Abstract

Few-layer stacked niobium selenide (NbSe₂) has evoked great interest owing to its intrinsically exotic properties and accessible manipulation by controlled ion intercalation for superconductivity physics and advanced device applications. However, attempts to extend the range of reversible intercalation stoichiometries are often hindered by overexpanded bond rupture and intrinsic-limit transition metal redox centres in selenides when proceeding towards deep intercalation. Here, we report that reversible unconventional

nature portfolio

Peer Review File

Superstoichiometric reversible and manipulable copper-ion intercalation in niobium selenide

Corresponding Author: Professor Daming Zhu

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Attachments originally included by the reviewers as part of their assessment can be found at the end of this file.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

In this paper, the authors have achieved unconventional superstoichiometric intercalation of two Cu ions in NbSe₂. The mechanism behind this unique phenomenon is attributed to the synergistic redox reactions of cations and anions in NbSe₂, which stabilize the chemical bonds and lattice structure, thereby avoiding the occurrence of conversion reactions. Superstoichiometric intercalation enables high-capacity and stable copper-ion batteries and is extendable to other electrochemical energy storage systems. The proposed mechanism has certain innovation and significance. However, the evidence for superstoichiometric intercalation and synergistic cation-anion redox remains insufficient. Some experimental data and results still need to be further provided. My concerns and comments are appended below:

1. An important innovation of this work is that the author proposed the unconventional intercalation of two copper ions in NbSe₂ which is very stunning. However, the characterization confirming the intercalation of the two copper ions is semi-quantitative method, such as EDS, and ICP. I do not think the presented methods are sufficient to prove the intercalation of two copper ions. Therefore, the authors need to provide direct characterization methods, such as STEM, to atomically confirm the intercalation of two copper ions.
2. Exfoliating b-NbSe₂ to a few-layer f-NbSe₂, the amount of copper ion intercalation in NbSe₂ has been elevated to a superstoichiometric intercalation. The fundamental changes in f-NbSe₂ that lead to this phenomenon should be a key focus. However, the authors have almost ignored the theoretical calculations, electronic structure, and compositional changes of b-NbSe₂. Therefore, it is recommended that the authors supplement or add relevant data to compare the fundamental changes occurring in these two materials.

Anonymous reviews and author responses are given as supplemental material

Public, signed reviews

[Home](#) » [Browse](#) » What is open peer review? A systematic review

SYSTEMATIC REVIEW



REVISED What is open peer review? A systematic review
[version 2; peer review: 4 approved]

 Tony Ross-Hellauer 

 Author details

 This article is included in the [Research on Research, Policy & Culture](#) gateway.

Abstract

Background: “Open peer review” (OPR), despite being a major pillar of Open Science, has neither a standardized definition nor an agreed schema of its features and implementations. The literature reflects this, with numerous overlapping and contradictory definitions. While for some the term refers to peer review where the identities of both author and reviewer are disclosed to each other, for others it signifies systems where reviewer reports are published alongside articles. For others it signifies both of these conditions, and for yet others it describes systems where not only “invited experts” are able to comment. For still others, it includes a variety of combinations of these and other novel methods.

Methods: Recognising the absence of a consensus view on what open peer review is, this article undertakes a systematic review of definitions of “open peer review” or “open review”, to create a corpus of 122 definitions. These definitions are systematically analysed to build a coherent typology of the various innovations in peer review signified by the term, and hence provide the precise technical definition currently lacking.

Results: This quantifiable data yields rich information on the range and extent of differing definitions over time and by broad subject area. Quantifying definitions in this way allows us to accurately portray exactly how ambiguously the phrase “open peer review” has been used thus far, for the literature offers 22 distinct configurations of seven traits, effectively meaning that there are 22 different definitions of OPR in the literature reviewed.

Conclusions: I propose a pragmatic definition of open peer review as an umbrella term for a number of overlapping ways that peer review models can be adapted in line with the aims of Open Science, including making reviewer and author identities open, publishing review reports and enabling greater participation in the peer review process.

Metric	Value
Views	56729
Downloads	5658

 [Get PDF](#) [Get XML](#)

“Cite

Export

 Track Share

[← Back to all reports](#)

Reviewer Report

371 Views ©

22 May 2017 | for Version 1

Emily Ford , Urban & Public Affairs Librarian, Portland State University, Portland, OR, USA

” Cite this report

Responses (1)

? APPROVED WITH RESERVATIONS

Introduction

- The definition of open science needs to be clearly stated in the Introduction in order to strengthen the frame of the whole paper. Is the definition you are using of open science fully accepted and not contested? If so, then great, but if not, then it becomes murkier and you might want to spend time unpacking the tension there. Also in the last sentence of the Intro, what is that ethos of open science?

Introduction: Background

- Would it be useful to unpack some counter arguments on the reasons peer review in its current state of

F1000Research

Search

BROWSEGATEWAYS & COLLECTIONSHOW TO PUBLISHABOUT

Home » Browse Articles

ArticlesFaculty ReviewsDocumentsPostersSlides

FILTERS21-40 of 10,045 ARTICLES

RESEARCH ARTICLE metrics

Machine learning applications in risk management: Trends and research agenda

[version 1; peer review: 1 approved with reservations, 1 not approved]

Alejandro Valencia-Arias, Jesus Alberto Jimenez Garcia, Erica Agudelo-Ceballos, Aarón José Alberto Oré León, Ezequiel Martínez Rojas, Julio Leyrer Henríquez, Diana Marleny Ramírez-Ramírez

PEER REVIEWERS Ajay Vikram Singh; Rakibul Hasan Chowdhury

PUBLISHED 25 Feb 2025

RESEARCH ARTICLE metricsAWAITING PEER REVIEW

Syphilis in pregnancy: Prevalence, risk factors and maternal-fetal prognosis in Kisangani

[version 1; peer review: awaiting peer review]

Likilo Osundja Jérémie, Tshodi Bulanda Arsène, Lufuluabu Mpemba Alphonse, Mbohhou Kouaming Aminate Nawal, Komanda Likeweke Emmanuel, Katenga Bosunga Gédéon

PEER REVIEWERS Invited

FUNDER none

PUBLISHED 25 Feb 2025

RESEARCH ARTICLE metrics

REVISED Sense of personal agency towards mitigating the threat of antibiotic resistance: a focus group study with parents of children under 5 years old, conducted mid-pandemic

[version 2; peer review: 2 approved with reservations]

Becky McCall, Andrew Hayward, Michael Wilson, Gill Forbes, Laura Shallcross

F1000Research

Search

BROWSEGATEWAYS & COLLECTIONSHOW TO PUBLISHABOUT

MY RESEARCHSIGN IN

SUBMIT YOUR RESEARCH

Home » Browse » Through restful waters and deep commotion: A study on burnout and...

Check for updates

RESEARCH ARTICLE

REVISED Through restful waters and deep commotion: A study on burnout and health impairment of Italian seafarers from the JD-R model perspective. [version 2; peer review: 3 approved]

Francesco Buscema1, Lorenzo Cena1, Clarissa Cricenti2, Margherita Zito3, Lara Bertola4, Lara Colombo1

Author details

 This article is included in the Social Psychology gateway.

ALL METRICS

264 VIEWS

25 DOWNLOADS

Get PDF

Get XML

Cite

Export

Track

Share

Abstract

Background

The work experience of seafarers differs significantly from other land-based occupations due to several factors, particularly remoteness and the restricted work environment. This study seeks to examine the impact of burnout and health impairment in the maritime industry, using the Job Demand-Resources theory as a framework.

Methods

To investigate these phenomena, an online questionnaire was sent to 629 Italian seafarers and we conducted analysis on a valid sample of 239 respondents (94.6% men, Mage = 39.44, SD = 12.8). We tested a mediated Structural Equation Model (SEM) aimed at predicting negative health outcomes.

Results

The results show that burnout plays a mediating role between job demands (such as workload and cognitive strain) and health impairment (such as sleep quality and physical well-being) (Total Indirect Effect = 0.443, $p < .0001$) as well as between job resources (such as social support and transformational leadership) and health impairment (Total Indirect Effect = -0.249, $p < .0001$). Furthermore, the findings highlight the direct influence of occupational resources on seafarers' health.

Conclusions

The discussion highlights the urgent need for more research in the field of organisational psychology in the maritime industry and the discrepancies between these findings, which are consistent with the existing maritime literature, and other studies that do not include seafarers in their sample groups.

Open Peer Review

Reviewer Status

Reviewer Reports

	Invited Reviewers		
	1	2	3
Version 2 (revision) 20 Feb 25	✓ read	✓ read	
Version 1 03 Jan 25	? read	? read	✓ read

1. Inga Bartuseviciene, World Maritime University, Malmö, Sweden

2. Vairavan C, AMET Deemed to be University, Chennai, India

3. Andrea Russo, University of Split, Split, Croatia

Comments on this article

All Comments (0)

Add a comment

Sign up for content alerts

Email

SIGN UP

Browse by related subjects

Alternative publication models

- A brief history of scholarly publishing and reviewing models
- Consequences of paper-based publishing models
- What is our reviewing model?
- Alternative reviewing models
- **Alternative publication models**
- A more radical reviewing and publication model
- Potential benefits of new models
- Concluding thoughts

General Relativity and Quantum Cosmology

[Submitted on 5 Oct 2017 (v1), last revised 8 Oct 2019 (this version, v4)]

Effects of Data Quality Vetoes on a Search for Compact Binary Coalescences in Advanced LIGO's First Observing Run

The [LIGO Scientific Collaboration](#), the [Virgo Collaboration](#): B. P. Abbott, R. Abbott, T. D. Abbott, M. R. Abernathy, F. Acernese, K. Ackley, C. Adams, T. Adams, P. Addesso, R. X. Adhikari, V. B. Adya, C. Affeldt, M. Agathos, K. Agatsuma, N. Aggarwal, O. D. Aguiar, L. Aiello, A. Ain, B. Allen, A. Allocca, P. A. Altin, S. B. Anderson, W. G. Anderson, K. Arai, M. C. Araya, C. C. Arceneaux, J. S. Areeda, N. Arnaud, K. G. Arun, S. Ascenzi, G. Ashton, M. Ast, S. M. Aston, P. Astone, P. Aufmuth, C. Aulbert, S. Babak, P. Bacon, M. K. M. Bader, P. T. Baker, F. Baldaccini, G. Ballardín, S. W. Ballmer, J. C. Barayoga, S. E. Barclay, B. C. Barish, D. Barker, F. Barone, B. Barr, L. Barsotti, M. Barsuglia, D. Barta, J. Bartlett, I. Bartos, R. Bassiri, A. Basti, J. C. Batch, C. Baune, V. Bavigadda, M. Bazzan, M. Bejger, A. S. Bell, B. K. Berger, G. Bergmann, C. P. L. Berry, D. Bersanetti, A. Bertolini, J. Betzwieser, S. Bhagwat, R. Bhandare, I. A. Bilenko, G. Billingsley, J. Birch, R. Birney, S. Biscans, A. Bisht, M. Bitossi, C. Biwer, M. A. Bizouard, J. K. Blackburn, C. D. Blair, D. G. Blair, R. M. Blair, S. Bloemen, O. Bock, M. Boer, G. Bogaert, C. Bogan, A. Bohe, C. Bond, F. Bondu, R. Bonnand, B. A. Boom, R. Bork, V. Boschi, S. Bose, Y. Bouffanais, A. Bozzi et al. (860 additional authors not shown)

The first observing run of Advanced LIGO spanned 4 months, from September 12, 2015 to January 19, 2016, during which gravitational waves were directly detected from two binary black hole systems, namely GW150914 and GW151226. Confident detection of gravitational waves requires an understanding of instrumental transients and artifacts that can reduce the sensitivity of a search. Studies of the quality of the detector data yield insights into the cause of instrumental artifacts and data quality vetoes specific to a search are produced to mitigate the effects of problematic data. In this paper, the systematic removal of noisy data from analysis time is shown to improve the sensitivity of searches for compact binary coalescences. The output of the PyCBC pipeline, which is a python-based code package used to search for gravitational wave signals from compact binary coalescences, is used as a metric for improvement. GW150914 was a loud enough signal that removing noisy data did not improve its significance. However, the removal of data with excess noise decreased the false alarm rate of GW151226 by more than two orders of magnitude, from 1 in 770 years to less than 1 in 186000 years.

Comments: 27 pages, 13 figures, published version
Subjects: **General Relativity and Quantum Cosmology (gr-qc)**; Instrumentation and Methods for Astrophysics (astro-ph.IM)
Cite as: [arXiv:1710.02185 \[gr-qc\]](#)
(or [arXiv:1710.02185v4 \[gr-qc\]](#) for this version)
<https://doi.org/10.48550/arXiv.1710.02185> ⓘ
Journal reference: Class. Quantum Grav. 35 065010 (2018)
Related DOL: <https://doi.org/10.1088/1361-6382/aaaafa> ⓘ

Submission history
From: LVC Publications [[view email](#)] [via Lvc Publications as proxy]
[\[v1\]](#) Thu, 5 Oct 2017 19:18:51 UTC (4,609 KB)
[\[v2\]](#) Wed, 11 Oct 2017 16:26:26 UTC (4,608 KB)
[\[v3\]](#) Tue, 7 Nov 2017 22:45:22 UTC (4,608 KB)
[\[v4\]](#) Tue, 8 Oct 2019 13:29:33 UTC (4,507 KB)

Access Paper:

- [View PDF](#)
 - [TeX Source](#)
 - [Other Formats](#)
- view license

Current browse context:
gr-qc
< prev | next >
[new](#) | [recent](#) | [2017-10](#)
Change to browse by:
[astro-ph](#)
[astro-ph.IM](#)

References & Citations

- [INSPIRE HEP](#)
- [NASA ADS](#)
- [Google Scholar](#)
- [Semantic Scholar](#)

Export BibTeX Citation

Bookmark



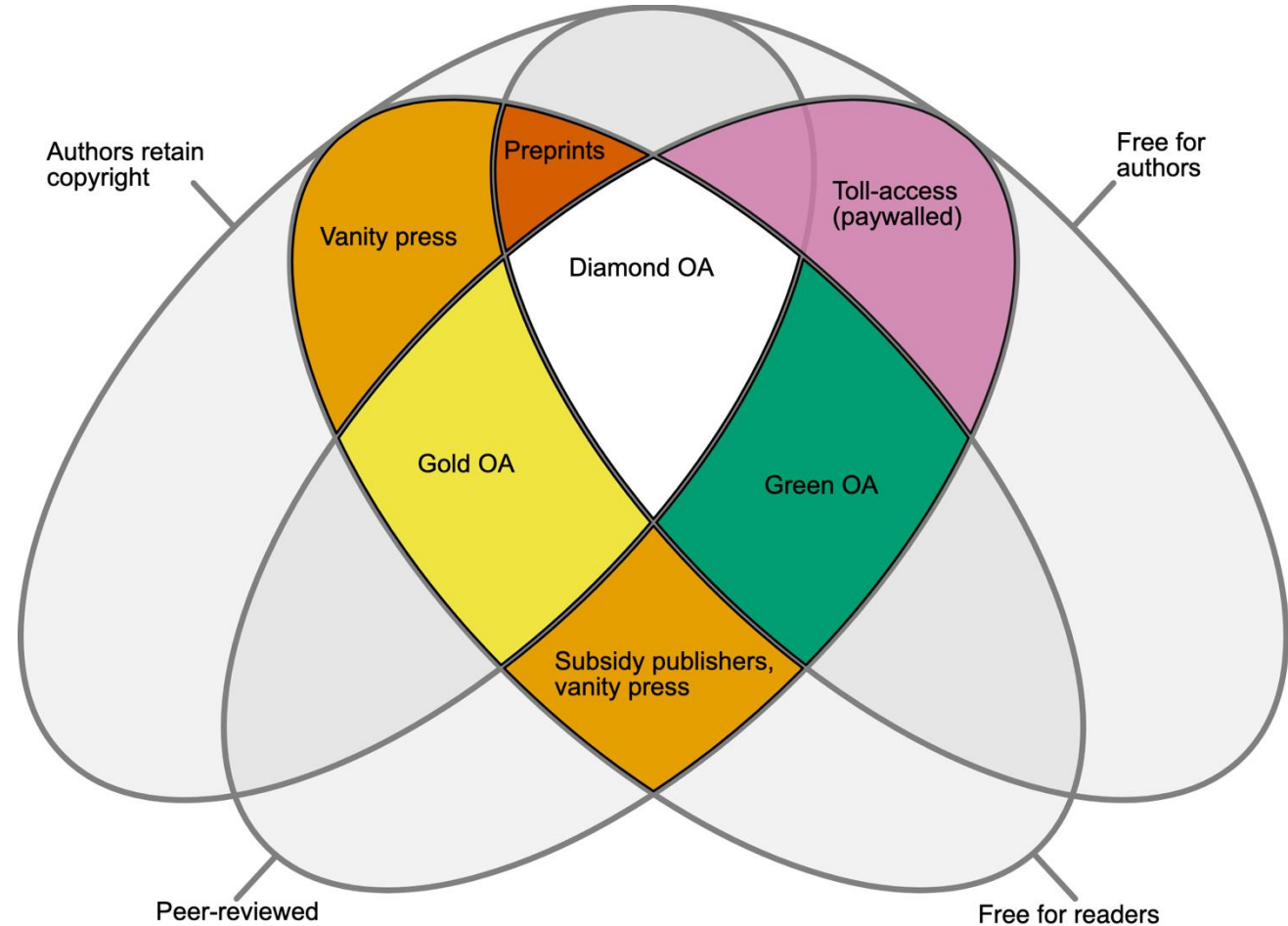
- Bibliographic Tools
- Code, Data, Media
- Demos
- Related Papers
- About arXivLabs

Bibliographic and Citation Tools

- ☐ Bibliographic Explorer ([What is the Explorer?](#))
- ☐ Connected Papers ([What is Connected Papers?](#))
- ☐ Litmaps ([What is Litmaps?](#))
- ☐ scite Smart Citations ([What are Smart Citations?](#))

Open Access

- Articles require no fee or subscription to read
- Authors pay fee to publish
 - Except for Diamond / Green OA, free to publish and read
- Obvious moral hazard
 - Predatory journals solicit articles for publication fee, but don't provide quality such as careful reviewing
- Open Access most widespread Open Science principle

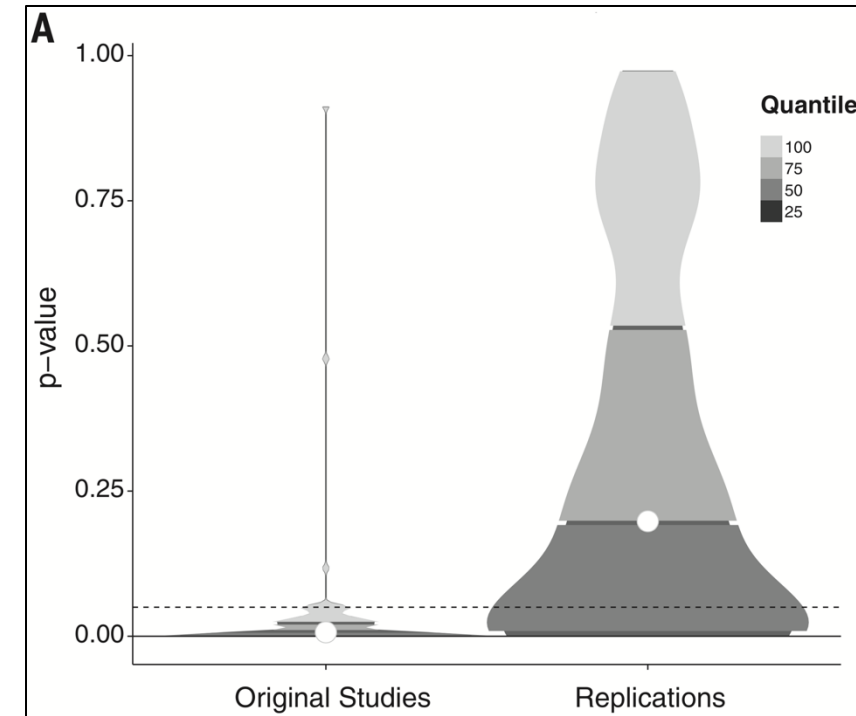


A more radical reviewing and publication model

- A brief history of scholarly publishing and reviewing models
- Consequences of paper-based publishing models
- What is our reviewing model?
- Alternative reviewing models
- Alternative publication models
- A more radical reviewing and publication model
- Potential benefits of new models
- Concluding thoughts

Motivation: preregistered empirical plans

- Before collecting empirical data, create detailed, written plan
 - Hypothesis, methods, analysis
- Removes possibility of ***p*-hacking**
 - Only significant ($p \leq 0.05$) results **analyzed**
 - Only significant ($p \leq 0.05$) results **published**
 - Only significant ($p \leq 0.05$) results **reported**
- P-hacking results in **irreproducible research**
 - **Original studies**: *p*-hacking likely
 - **Replicated studies**: all results reported, no *p*-hacking
- **Best practice**: publicly preregister the plan
 - Center for Open Science (<https://www.cos.io/>)



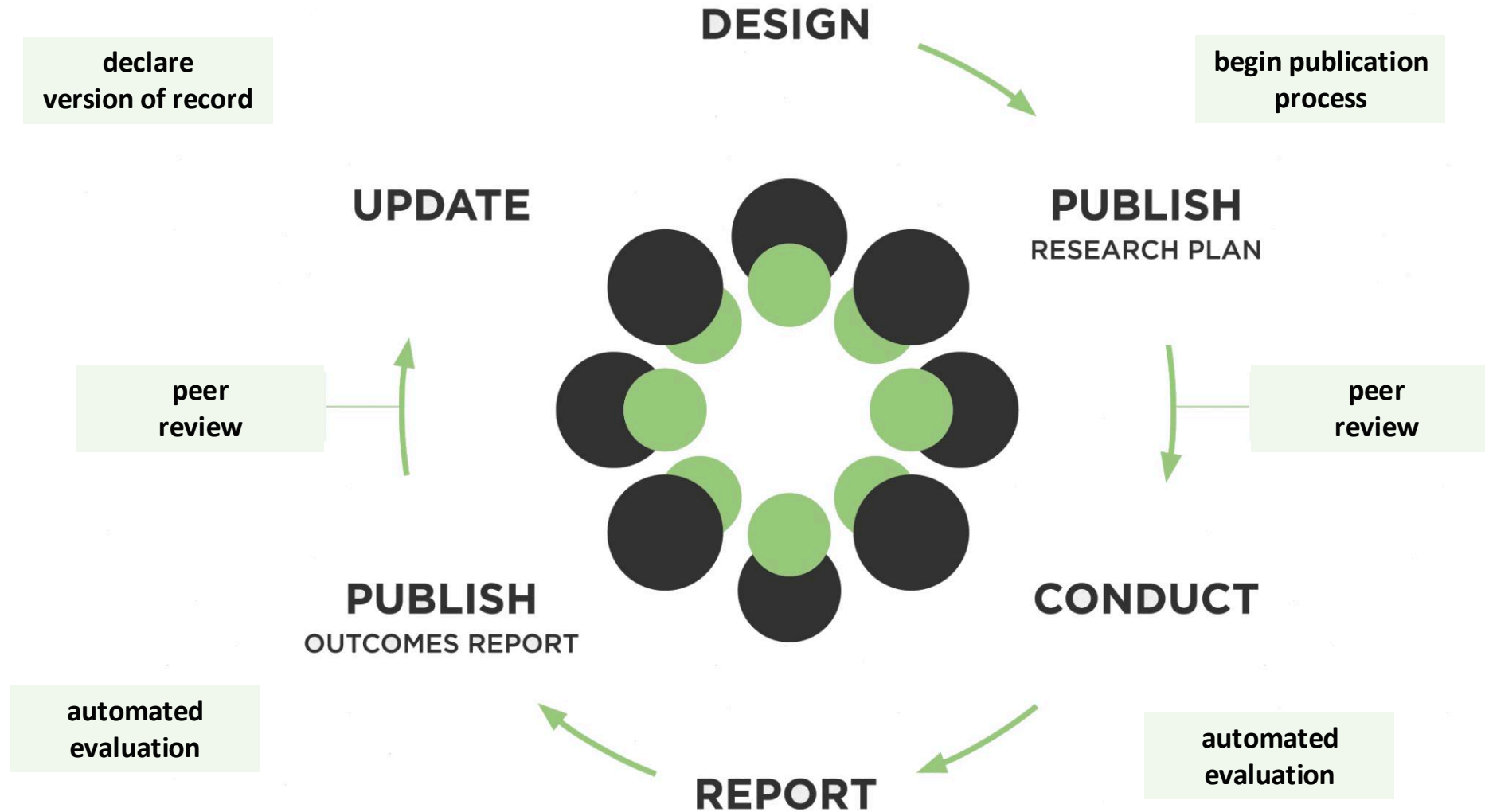
Reproducibility Project: Psychology
 $p \leq 0.05$ all 100 original studies
 $p \leq 0.05$ for 36 replications

What if pre-registered plans are peer reviewed?



- If pre-registered plans are peer reviewed (Stage 1), then:
 - Reviewers agree that, **regardless of outcome**, the empirical study **should be conducted**
 - Stage 1 reviews change reviewing context of final paper (Stage 2)
- Because results at Stage 1 not yet known → bias against reporting negative results eliminated
- Therefore, author incentives change:
producing **publishable story** → producing **accurate story**

Generalize Idea: Lifecycle Journal



Reviewing and evaluation services



human peer review



ResearchHub



AI / automated review



DataSeer



Shared evolutionary origins of nectarines, sauerkraut and Mr. Bean.

Smith, Smyth, and Smythe

Peer reviewed. *Lifecycle Journal*. Version 1, 1 May 2023; Version 4, 10 June 2024. DOI: [lj0001232.v4](#)

-  Experiment 1 preregistration
-  Experiment 1 materials
-  Experiment 1 data
-  Experiment 2 data
-  Paper

Abstract

In this paper, we delve into the multifaceted interplay between quantum phenomena and cognitive dynamics, elucidating the intricate connections that underpin their complex relationship. Through a synthesis of theoretical frameworks and empirical evidence, we endeavor to unravel the enigmatic synergies between these seemingly disparate domains. Employing a multidisciplinary approach, we traverse the landscape of quantum mechanics and cognitive science, traversing the realms of uncertainty and cognition. Our exploratory analysis navigates the subtleties of entanglement and perception, probing the boundaries of conventional understanding. By weaving together threads of quantum entanglement and cognitive processes, we illuminate novel perspectives that challenge traditional paradigms. Through this synthesis, we aspire to catalyze further inquiry into the profound intersections of quantum phenomena and cognitive dynamics, fostering a deeper appreciation for the inherent complexity of the universe and the mind.

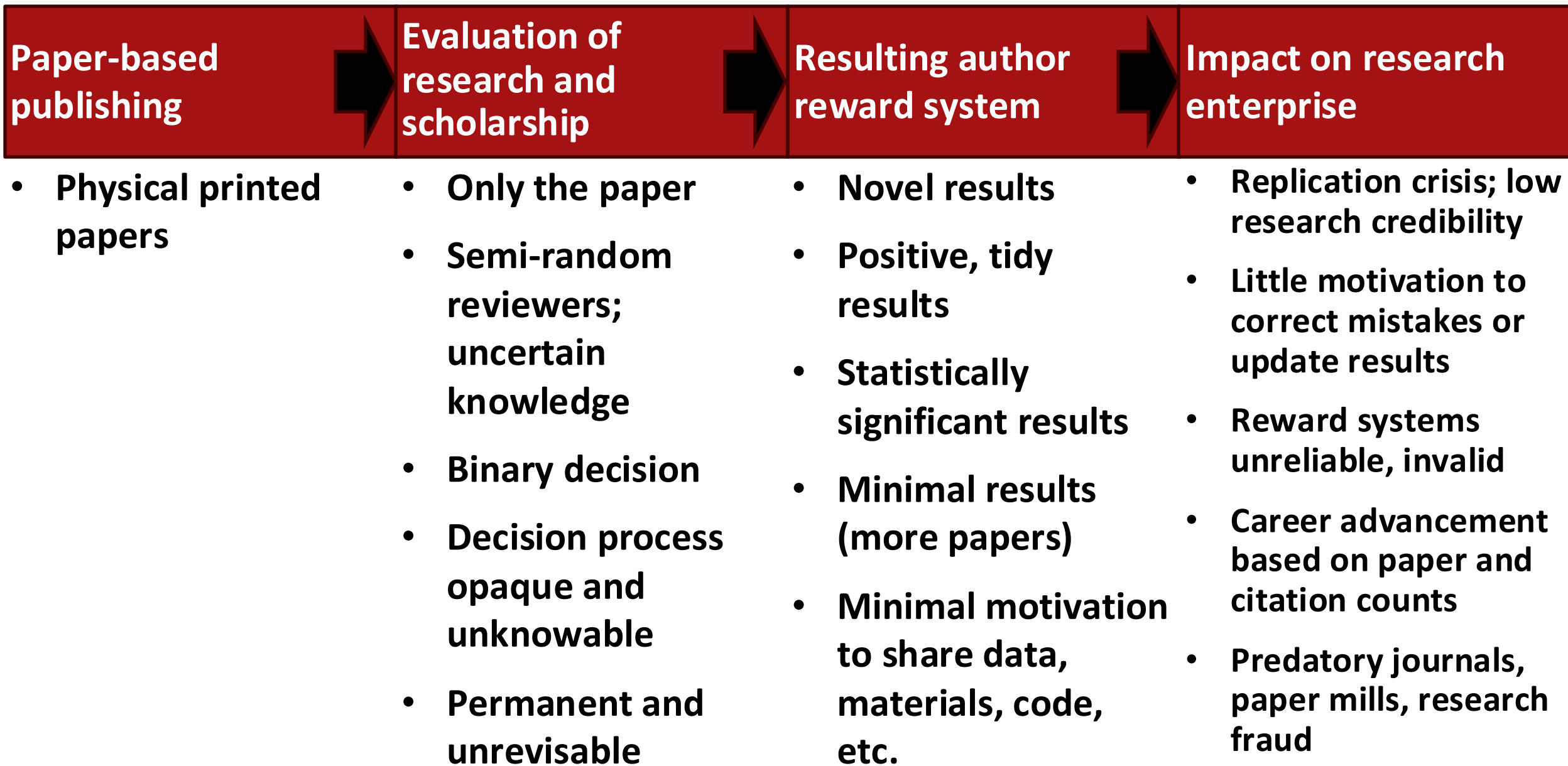
Service	Summary of most recent evaluation	Version	Most recent	Evaluated
	Recommended by three reviewers. [Read more]	2, 4	1 August 2024	
	Aggregate score = 75% likely to replicate. [Read more]	2	12 October 2023	 
	Found 2 minor errors, 1 major error. [Read more]	1	5 May 2023	   
 DataSeer	Found 3 of 4 data sets shared. [Read more]	1	4 May 2023	

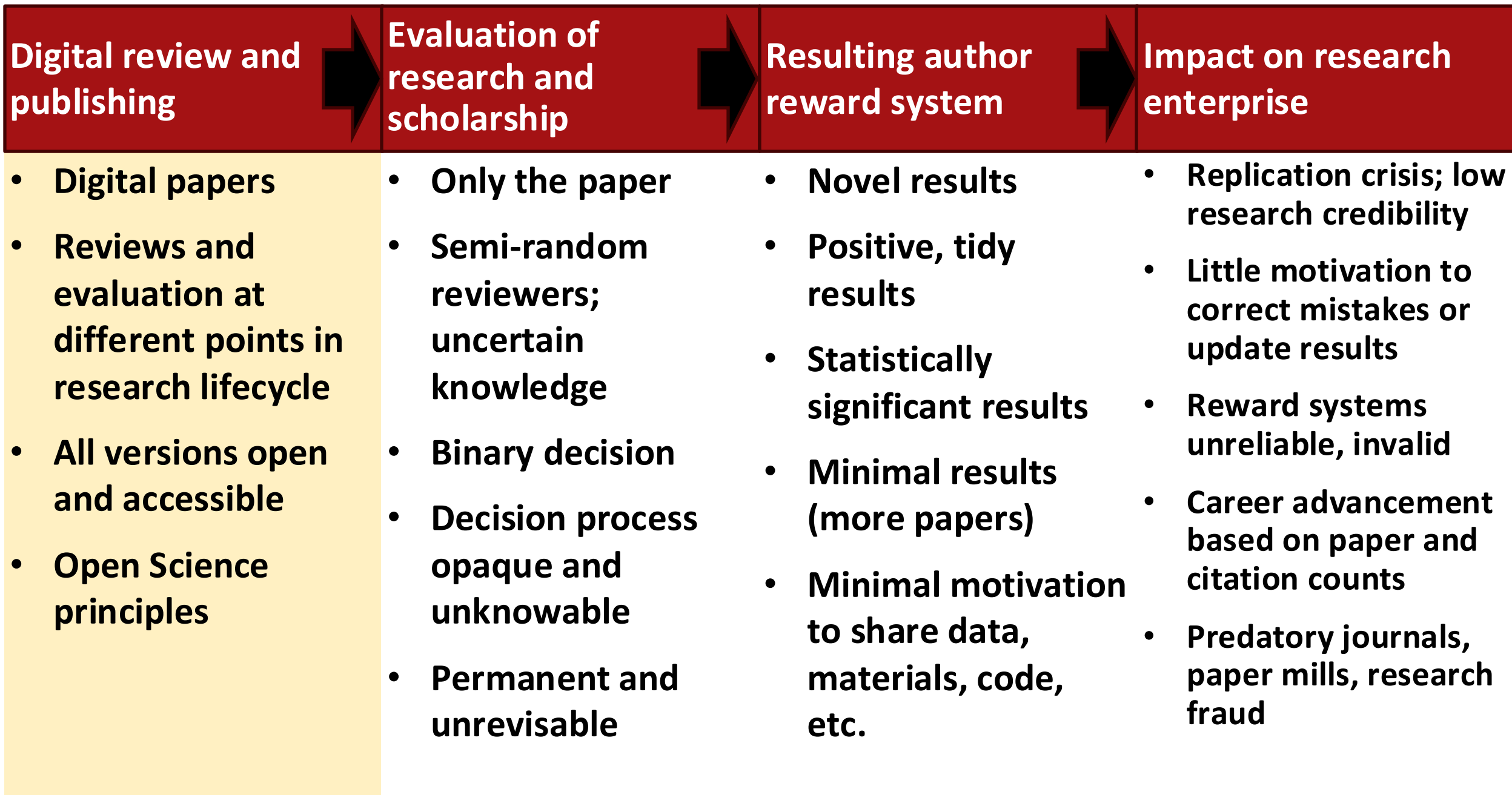
Lifecycle journal papers

- When is a paper finished?
 - Author assigns a version to be a Version of Record (can still update later)
 - All versions available
- Isn't this more work than current writing and publishing methods?
 - Perhaps, perhaps not.
 - If rigor of the work is increased → authors find effort more rewarding
 - If rigor of the work is increased → authors have increased reputational rewards
- How is it cited?
 - Smith, J. A., Smyth, K. B., & Smythe, L. C. (2024). Shared evolutionary origins of nectarines, sauerkraut and Mr. Bean. *Lifecycle Journal*. DOI: [lj0001232.v4](#)
 - Recommended by *Peer Community In: Registered Reports*
 - Reproduced by *Institute for Replication*
 - Analysis rated correct by *Statcheck*

Potential benefits of new models

- A brief history of scholarly publishing and reviewing models
- Consequences of paper-based publishing models
- What is our reviewing model?
- Alternative reviewing models
- Alternative publication models
- A more radical reviewing and publication model
- Potential benefits of new models
- Concluding thoughts





e.g., [Nosek, 2024]

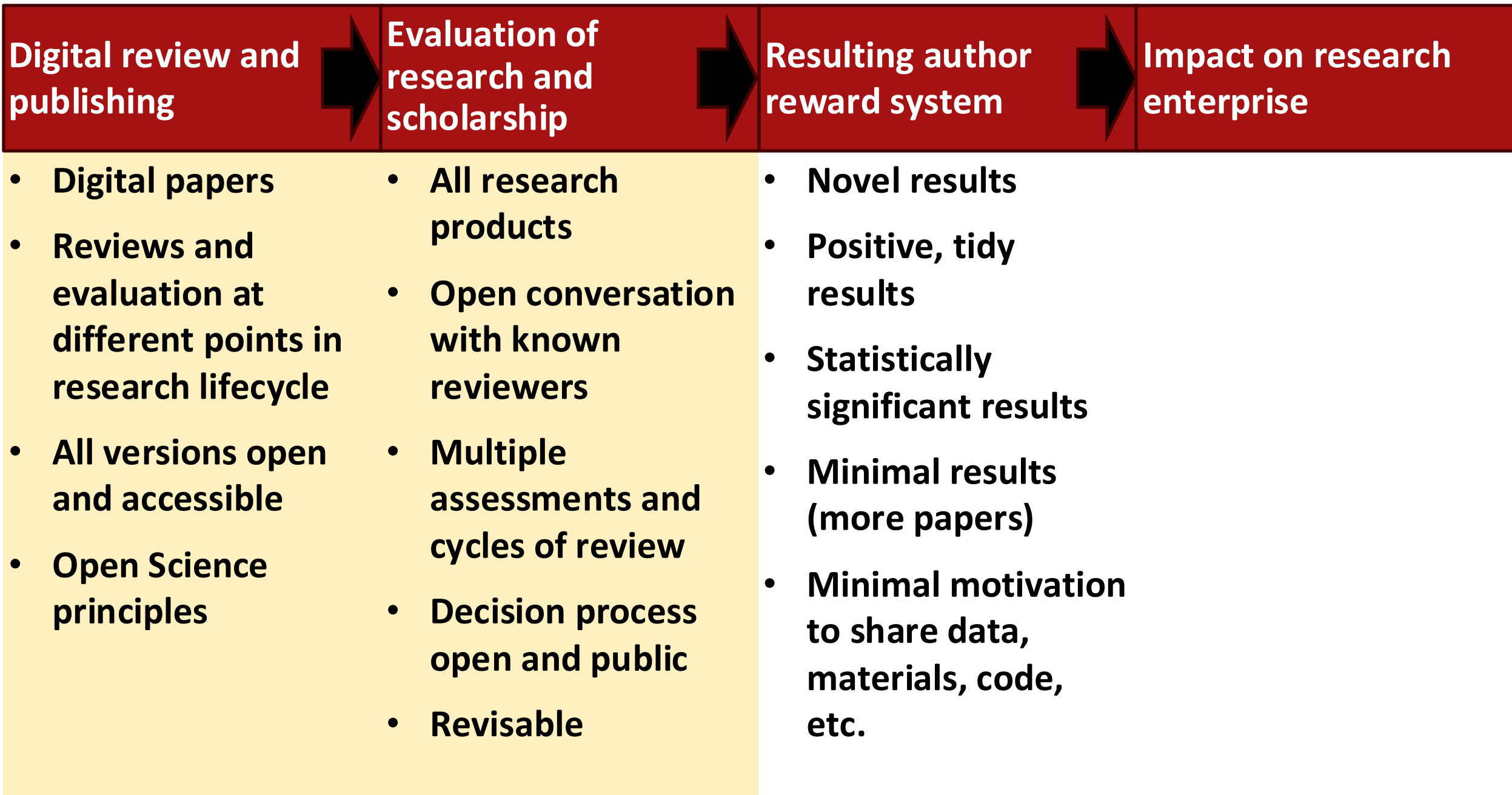


- Digital papers
- Reviews and evaluation at different points in research lifecycle
- All versions open and accessible
- Open Science principles

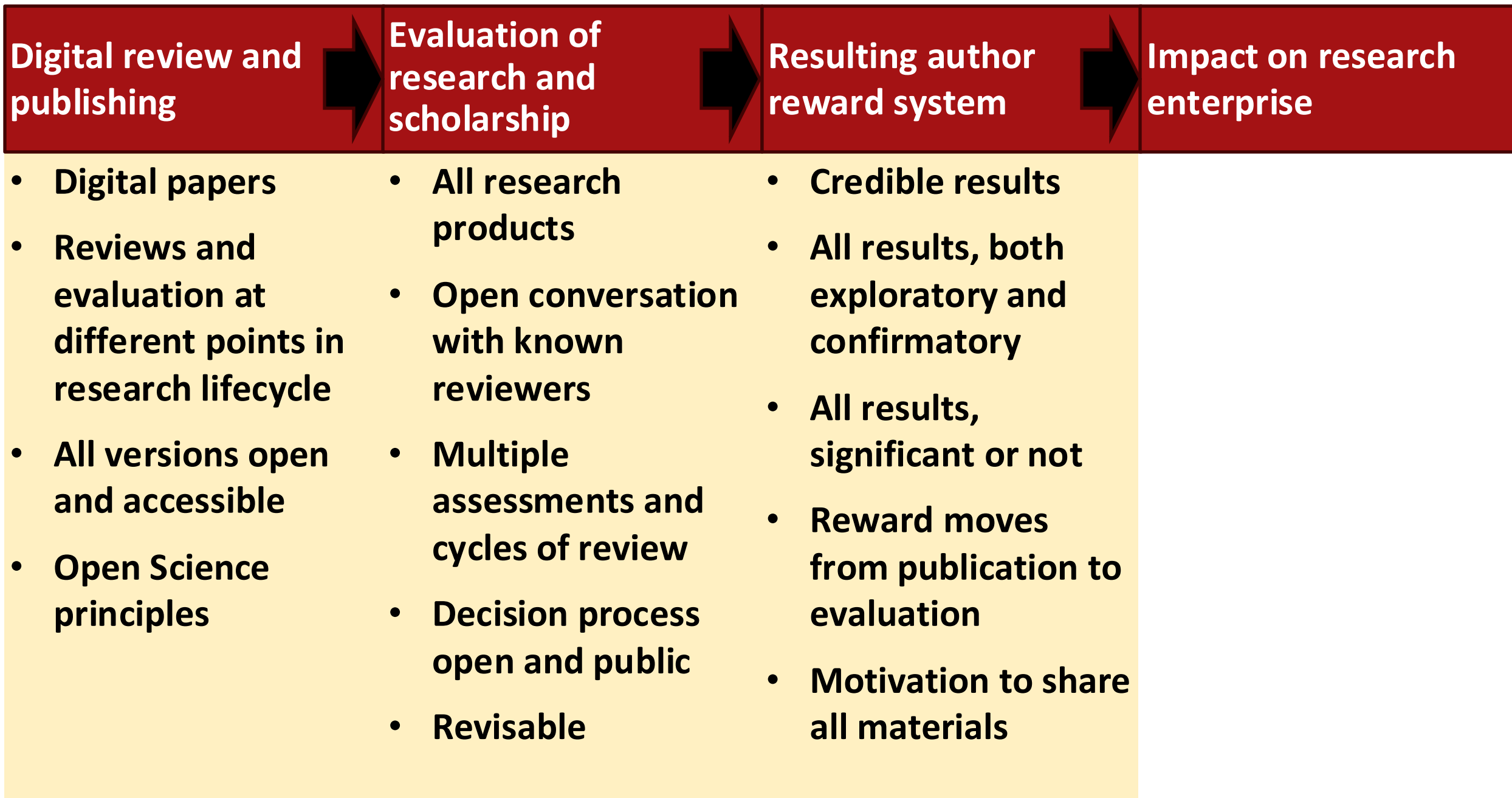
- Only the paper
- Semi-random reviewers; uncertain knowledge
- Binary decision
- Decision process opaque and unknowable
- Permanent and unrevisable

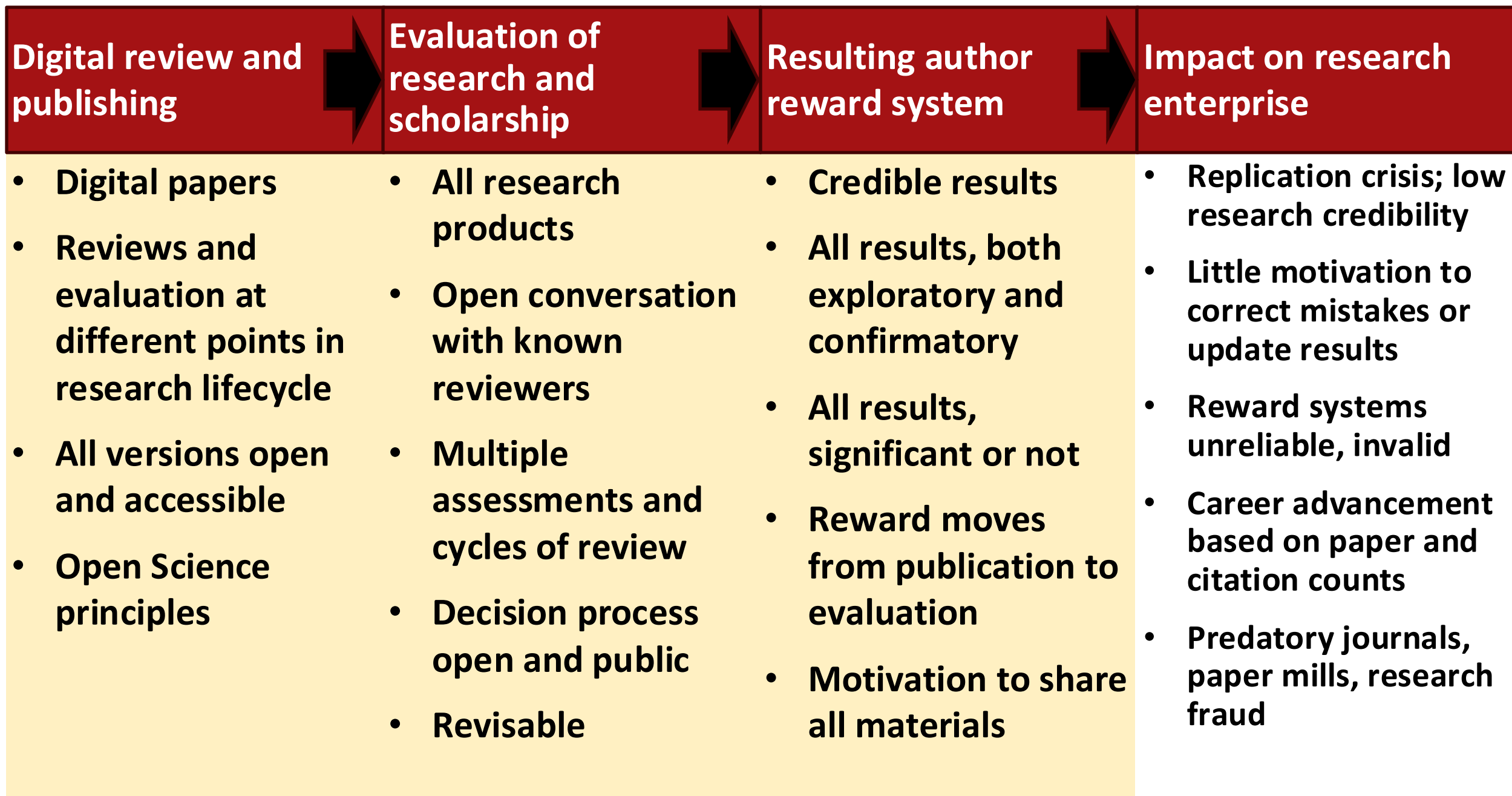


- Digital papers
 - Reviews and evaluation at different points in research lifecycle
 - All versions open and accessible
 - Open Science principles
- All research products
 - Open conversation with known reviewers
 - Multiple assessments and cycles of review
 - Decision process open and public
 - Revisable

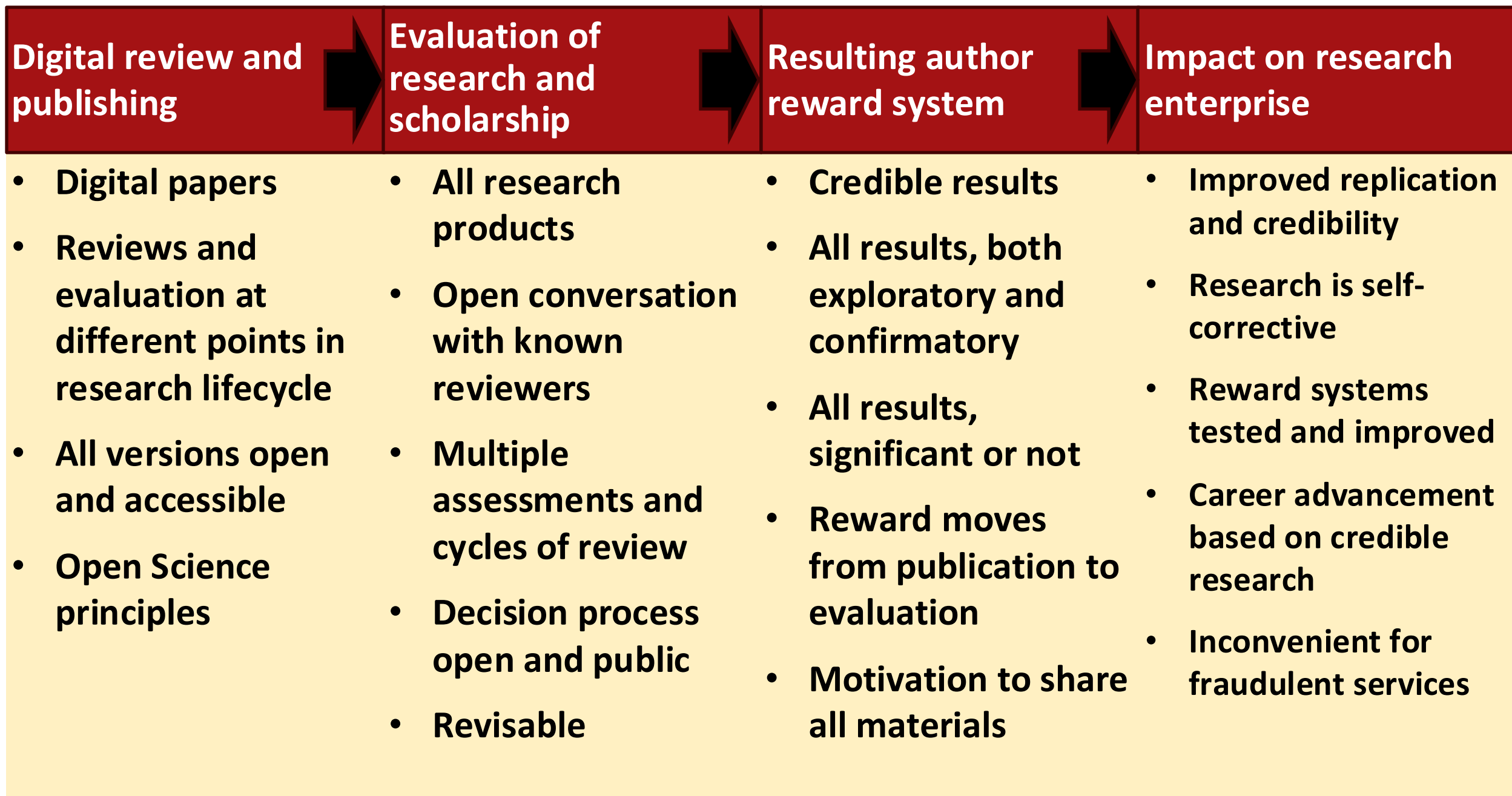


e.g., [Nosek, 2024]





e.g., [Nosek, 2024]



Concluding thoughts

- A brief history of scholarly publishing and reviewing models
- Consequences of paper-based publishing models
- What is our reviewing model?
- Alternative reviewing models
- Alternative publication models
- A more radical reviewing and publication model
- Potential benefits of new models
- **Concluding thoughts**

Why did you get into science and technology?

- Think back...what inspired you?
- Was it:
 - Applying for grants?
 - Fighting to get papers published in high impact-factor outlets?
 - Fighting for big h-index?
- Scholarly communication practices → scholarly rewards
 - Current practices misaligned with how knowledge actually produced
- Knowledge production is a **social process** that supports the dynamic exchange of:
 - Ideas
 - Evidence
 - Explanations
 - Identification of flaws
 - Exploration of alternatives
- **Reasons for optimism:**
 - There is an energetic scholarly community working to improve current practices
 - Potential of digital review and publishing to improve the process is not yet fully realized

References

[Fyfe, 2015] Laloup, J., “The history of scientific publishing: An interview with Aileen Fyfe”, *Podcast*,
<https://web.archive.org/web/20171002092042/https://www.timeshighereducation.com/features/peer-review-not-old-you-might-think>

[Hidden Brain, 2021] Hidden Brain Staff, “Our Noisy Minds: Interview with Daniel Kahneman”, *Hidden Brain*, Podcast, May 18, 2021

[Kahneman, Sibony, Sustein, 2021] D Kahneman, O Sibony, C Sunstein, *Noise: A flaw in human judgement*, Little, Brown Spark, 2021

[Nosek, 2024] Nosek, B., “Reimagining Scholarly Publishing to Promote Credible and Trustworthy Research”, Capstone talk given at *ACM Virtual Reality Science and Technology (VRST) 2024*, <https://osf.io/c586f>

[Open Science Collaboration 2015] Open Science Collaboration, “Estimating the Reproducibility of Psychological Science”, *Science*, 349(6251), 2015, DOI: 10.1126/science.aac4716

[Open Science Collaboration 2012] Open Science Collaboration, “An Open, Large-Scale, Collaborative Effort to Estimate the Reproducibility of Psychological Science”, *Perspectives on Psychological Science*, 7(6), 657–660, 2012.
<http://doi.org/10.1177/1745691612462588>

[Surowiecki, 2005] J Surowiecki, *The wisdom of crowds*, Vintage, 2005

https://en.wikipedia.org/wiki/Scholarly_peer_review

<https://www.aje.com/arc/scholarly-publishing-brief-history/>

<https://www.britannica.com/topic/publishing/Scholarly-journals>

https://en.wikipedia.org/wiki/Academic_publishing

https://en.wikipedia.org/wiki/Shadow_library

<https://www.computer.org/publications/author-resources>,
Double-Anonymous Review

https://en.wikipedia.org/wiki/Diamond_open_access

https://en.wikipedia.org/wiki/Open_access

<https://www.cos.io/>

<https://www.cos.io/initiatives/registered-reports>

<https://lifecyclejournal.org/>

Contact Information

J. Edward Swan II

Department of Computer Science and Engineering

Mississippi State University

swan@acm.org

Slide Location:

ed-swan.github.io/teaching/tutorials/Swan-VR2025-Reviewing-Publication-Models.pdf

IEEE VR



SAINT-MALO, FRANCE
March 8-12, 2025

Reviewing and Publication Models in the Era of Online Publishing and Generative AI

J. Edward Swan II

Mississippi State University

Saturday, 8 March 2025